The third magnetic Neumann eigenvalue does not exceed the Dirichlet ground state on an annulus

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Abstract

Let $\mathcal{A}_a = \{x \in \mathbb{R}^2 : a < |x| < 1\}$, $0 < a < 1$, and consider the two-dimensional magnetic Laplacian $(-i\nabla - \mathbf{A}_b)^2$ generated by the homogeneous field in symmetric gauge, $\mathbf{A}_b = b(-x_2, x_1)/2$, $b > 0$. Both boundary circles carry either magnetic Neumann or Dirichlet conditions, and no independent Aharonov–Bohm flux is added. We prove that the third Neumann eigenvalue, counted with multiplicity, never exceeds the Dirichlet ground-state energy. Thus the additional boundary component and the associated rearrangement of angular momenta do not destroy the third-Neumann/first-Dirichlet ordering known numerically for the disk. The proof is exact and partitions the field–radius plane into three regimes. For small fields, a quotient-monotonicity argument transfers three disk-type trials to every annulus. Intermediate fields are controlled by restricting two explicit disk modes at the exactly solvable field $b = 2$. At high field, Landau modes settle flux windows containing three integers; the complementary windows are handled by moment bounds for thick annuli and transverse Dirichlet confinement for thin annuli. No finite angular cutoff, perturbative truncation, or numerical eigenvalue enclosure enters the proof.

Keywords: magnetic Laplacian; annulus; Neumann–Dirichlet eigenvalue comparison; homogeneous magnetic field; min–max principle
Mathematics Subject Classification: 35P15; 81Q10

1 Introduction

Comparison inequalities between Dirichlet and Neumann spectra are sensitive to geometry, topology, and magnetic flux. For the homogeneous magnetic Laplacian on a bounded convex planar domain, Lotoreichik proved $\mu_{k+1} \le \lambda_k$ for every k [4]. The same paper analyzed the unit disk by angular fibers: at the first Dirichlet threshold it obtained an analytic comparison for $b > 4$, numerical evidence for the remaining fields, and an exact crossing at $b = 2$. Earlier work of Frank and Laptev gives a one-level counting advantage at Landau thresholds on general domains [3]. These results do not directly give a two-level shift at an arbitrary Dirichlet threshold on an annulus, which is nonconvex and has two boundary components.

The topology also forces a convention that is immaterial on simply connected domains: adding a closed, non-exact one-form changes the flux and can change the spectrum [1]. Throughout this article we fix the standard symmetric potential for the homogeneous field and add no independent harmonic Aharonov–Bohm component. General background on the constant-field magnetic Laplacian can be found in [2].

Our result establishes the following annular comparison in this fixed model.

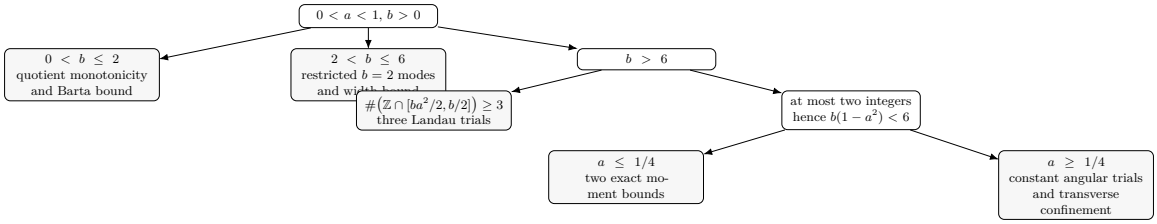


Figure 1: Exhaustive case partition used in the proof. The diagram records logical branches and exact estimates, not numerical data. It was rendered directly in LaTeX with the generative-AI assistance disclosed at the end of the article.

Theorem 1.1. *For every $0 < a < 1$ and $b > 0$, the globally ordered eigenvalues of the homogeneous-field magnetic Laplacian on $\mathcal{A}_a$, counted with multiplicity, satisfy*

$$\mu_3(\mathcal{A}_a, b) \le \lambda_1(\mathcal{A}_a, b),$$

where μ_j denotes the spectrum with magnetic Neumann conditions on both circles and λ_j the spectrum with Dirichlet conditions on both circles.

The proof is variational. It constructs three mutually orthogonal angular trials below explicit lower bounds for the Dirichlet ground state. The choice of trials changes at $b = 2$ and $b = 6$, and at high field it also depends on the number of integers in the annular flux window. Figure 1 records the exhaustive logical partition; the estimates themselves are proved in Sections 3–6.

In the chosen dimensionless normalization, the magnetic flux through the annulus divided by 2π is $b(1 - a^2)/2$. The theorem is therefore a spectral statement about an idealized one-particle model. It is not an assertion about a measured mesoscopic ring, a many-body system, or an added Aharonov–Bohm flux.

2 Operator, fibers, and a min–max criterion

Let

$$\mathcal{A}_a = \{(r\cos\theta, r\sin\theta) : a < r < 1\}, \qquad \mathbf{A}_b(x) = \frac{b}{2}(-x_2, x_1),$$

with $0 < a < 1$ and $b > 0$. The Neumann and Dirichlet realizations of

$$H_{a,b} = (-i\nabla - \mathbf{A}_b)^2$$

are associated with the same quadratic expression

$$q_{a,b}[u] = \int_{\mathcal{A}_a} |(-i\nabla - \mathbf{A}_b)u|^2 \,\mathrm{d}x$$

on $H^1(\mathcal{A}_a)$ and $H^1_0(\mathcal{A}_a)$, respectively. Because $\mathbf{A}_b$ is tangent to both boundary circles, the magnetic Neumann condition is the ordinary radial Neumann condition in every angular fiber.

For $u(r, \theta) = f(r)e^{im\theta}$, $m \in \mathbb{Z}$, the radial expression in $L^2((a, 1); r\,\mathrm{d}r)$ is

$$L_{m,b}f = -f'' - \frac{1}{r}f' + \left(\frac{m}{r} - \frac{br}{2}\right)^2 f. \tag{1}$$

The global spectra are the ordered union of these fibers, with degeneracies retained.

Lemma 2.1 (Three-trial criterion)**.** *Suppose $u_1, u_2, u_3 \in H^1(\mathcal{A}_a)$ have distinct angular indices and each has Rayleigh quotient at most L. Then $\mu_3(\mathcal{A}_a, b) \le L$. In particular, if $L \le \lambda_1(\mathcal{A}_a, b)$, the conclusion of Theorem 1.1 follows.*

Proof. Distinct angular sectors are orthogonal both in L^2 and for the magnetic quadratic form. Hence every nonzero vector in their three-dimensional span has quotient bounded by the largest of the three individual quotients. The min–max principle gives the claim. □

3 Three lower bounds for the Dirichlet ground state

We use complementary bounds adapted to low field, Landau windows, and thin rings.

Proposition 3.1. *For all $0 < a < 1$ and $b > 0$,*

$$\lambda_1(\mathcal{A}_a, b) \ge \begin{cases} 5 + b^2/4, & 0 < b \le 2, \\ 6, & b \ge 2, \end{cases} \tag{2.1}$$

$$\lambda_1(\mathcal{A}_a, b) \ge b, \tag{2.2}$$

$$\lambda_1(\mathcal{A}_a, b) \ge W(a) := \frac{\pi^2}{(1-a)^2} - \frac{1}{4a^2}. \tag{2.3}$$

Proof. Let $\mathbb{B}$ be the unit disk. Dirichlet domain monotonicity gives $\lambda_1(\mathcal{A}_a, b) \ge \lambda_1(\mathbb{B}, b)$. The Dirichlet disk ground state is radial for the homogeneous field [4, Section 5]. For the positive radial comparison function

$$\phi(r) = (1 - r^2)e^{-r^2/2}$$

one computes

$$\frac{L_{0,b}\phi}{\phi} = 6 + \frac{b^2 - 4}{4} r^2.$$

Barta's inequality, followed by minimization over $0 < r < 1$, proves (2.1).

For $f(a) = f(1) = 0$, direct completion of the square in any m-fiber gives

$$q^D_{m,a,b}[f] - b\|f\|^2 = 2\pi \int_a^1 r \left| f' - \left(\frac{m}{r} - \frac{br}{2} \right) f \right|^2 \mathrm{d}r \ge 0. \tag{2}$$

This proves (2.2) after summing the angular fibers.

Finally set $g = r^{1/2} f$. The endpoint conditions give

$$\int_a^1 r|f'|^2 \,\mathrm{d}r = \int_a^1 \left(|g'|^2 - \frac{|g|^2}{4r^2} \right) \mathrm{d}r \ge \left(\frac{\pi^2}{(1-a)^2} - \frac{1}{4a^2} \right) \int_a^1 |g|^2 \,\mathrm{d}r.$$

Dropping the nonnegative magnetic potential proves (2.3). □

4 Small fields: $0 < b \le 2$

Put $c = b/2 \in (0, 1]$ and $x = a^2$. Consider the three angular trials

$$1, \qquad re^{-cr^2/2}e^{i\theta}, \qquad re^{-cr^2/2}e^{-i\theta}. \tag{3}$$

Write their quotients as R_0, R_1, R_{-1}. Direct integration gives

$$R_0 = \frac{c^2(1+x)}{2} < 5 + c^2, \qquad R_{-1} = R_1 + 4c. \tag{4}$$

The $m = 1$ radial factor satisfies $L_{1,2c}f = 2cf$. If

$$I(x,c) = \int_a^1 r^3 e^{-cr^2}\,\mathrm{d}r = \frac{(1+cx)e^{-cx} - (1+c)e^{-c}}{2c^2},$$

integration by parts yields

$$R_1(x,c) = 2c + \frac{(1-c)e^{-c} - x(1-cx)e^{-cx}}{I(x,c)}. \tag{5}$$

At the disk endpoint $x = 0$ this implies

$$5 + c^2 - R_{-1}(0,c) = (1-c)\left(5 - c - \frac{2c^2}{e^c - 1 - c}\right). \tag{6}$$

For $0 < c \le 1$, positivity of the first four relevant terms in the exponential series gives

$$(5-c)(e^c - 1 - c) - 2c^2 \ge c^2\left(\frac{1}{2} + \frac{c}{3} - \frac{c^2}{6}\right) > 0. \tag{7}$$

Thus the disk gap in (6) is positive when $c < 1$.

It remains to show that removing a centered disk cannot increase R_1. Differentiation of (5) gives a positive prefactor times $P(x,c)$:

$$\partial_x R_1 = \frac{2c^2 e^c}{(cxe^c - ce^{cx} + e^c - e^{cx})^2}\,P(x,c), \tag{8}$$

where

$$\begin{aligned} P(x,c) = {} & c^3x^2e^{cx} - c^3xe^{cx} + c^2x^2e^c + c^2x^2e^{cx} - 2c^2xe^{cx} \\ & + 2cxe^c - 3cxe^{cx} + ce^{cx} - e^c + e^{cx}. \end{aligned}$$

Set $t = 1 - x$ and $H(t,c) = e^{ct-c}P(1-t,c)$. Exact differentiation gives

$$H(0,c) = H_t(0,c) = 0, \qquad H_{tt}(0,c) = c^2(c^2-1) \le 0, \tag{9}$$

$$H_{ttt}(t,c) = c^3e^{ct}\left[c^2(1-t)^2 - 4c(1-t) - 1\right] < 0 \tag{10}$$

for $0 < t \le 1$ and $0 < c \le 1$. Three integrations from $t = 0$ show $H(t,c) < 0$, hence $\partial_x R_1 < 0$ for $0 \le x < 1$.

For $c < 1$, equations (6)–(10) put both signed angular quotients strictly below $5 + c^2$. At $c = 1$ the disk gap vanishes, but the strict decrease for every annulus $a > 0$ gives the same conclusion. Together with (4), Proposition 3.1, and Lemma 2.1, this proves Theorem 1.1 for $0 < b \le 2$.

5 Intermediate fields: $2 \le b \le 6$

Use the constant and the two angular trials

$$f_1(r)e^{i\theta}, \quad f_1(r) = re^{-r^2/2}, \qquad f_2(r)e^{2i\theta}, \quad f_2(r) = r^2(3-r^2)e^{-r^2/2}. \tag{11}$$

At $b = 2$ these are exact disk eigenfunctions:

$$L_{1,2}f_1 = 2f_1, \qquad L_{2,2}f_2 = 6f_2, \qquad f_1'(1) = f_2'(1) = 0.$$

Moreover $f_j(a)f_j'(a) > 0$ for $0 < a < 1$. Restriction to the annulus and integration by parts therefore give

$$R_1(2) < 2, \qquad R_2(2) < 6. \tag{12}$$

Define the conditional moments

$$C_j(a) = \frac{\int_a^1 r^3 f_j(r)^2 \, \mathrm{d}r}{\int_a^1 r f_j(r)^2 \, \mathrm{d}r}, \qquad 0 < C_j(a) < 1.$$

Expanding the magnetic potential around $b = 2$ gives the exact shift

$$R_j(b) = R_j(2) + (b-2)\left[-j + \frac{b+2}{4} C_j(a)\right]. \tag{13}$$

Consequently,

$$R_1(b) < 2 + \frac{(b-2)^2}{4} \le 6, \qquad R_2(b) < 6 + \frac{(b-2)(b-6)}{4} \le 6. \tag{14}$$

The constant quotient is

$$R_0 = \frac{b^2(1+a^2)}{8}.$$

If $a^2 \le 1/3$, then $R_0 \le 6$. If $a^2 > 1/3$, then $a > 1/2$, $R_0 < 9$, and (2.3), with $\pi > 3$, gives $W(a) > 35$. In either case the three quotients are bounded by the Dirichlet ground-state lower bounds. Lemma 2.1 proves the theorem on this field interval.

6 High fields: $b > 6$

The high-field proof first uses exact Landau trials and then treats their integer-count complement.

6.1 The annular flux window

For $m \in \mathbb{Z}_{\ge 0}$ set

$$u_m(r,\theta) = r^m e^{-br^2/4} e^{im\theta}, \qquad I_{a,b,m} = \int_a^1 r^{2m+1} e^{-br^2/2} \, \mathrm{d}r.$$

The radial factor obeys

$$f_m' = \left(\frac{m}{r} - \frac{br}{2}\right) f_m, \qquad L_{m,b} f_m = b f_m.$$

Its exact Rayleigh quotient is

$$R_m = b + \frac{(m - b/2)e^{-b/2} - (m - ba^2/2)a^{2m}e^{-ba^2/2}}{I_{a,b,m}}. \tag{15}$$

Thus $R_m < b$ whenever

$$\frac{ba^2}{2} \le m \le \frac{b}{2}. \tag{16}$$

If the closed interval in (16) contains at least three integers, three orthogonal trials lie below $b \le \lambda_1$ by (2.2), and the proof is complete.

Suppose that the interval contains fewer than three integers. Every closed interval of length at least three contains at least three integers, so

$$b(1 - a^2) < 6. \tag{17}$$

6.2 The complement for $a \le 1/4$

The moments $C_j(a)$ from Section 5 increase with a: differentiating a truncated weighted mean shows that $C_j'(a)$ has the sign of $C_j(a) - a^2 > 0$. At $a = 1/4$, the substitution $z = r^2$ gives the exact identities

$$e\int_{1/16}^{1}(4/5 - z)ze^{-z}\,\mathrm{d}z = \frac{17}{5} - \frac{1637}{1280}e^{15/16}, \tag{18}$$

$$e\int_{1/16}^{1}(4/5 - z)z^2(3 - z)^2e^{-z}\,\mathrm{d}z = \frac{344}{5} - \frac{140644469}{5242880}e^{15/16}. \tag{19}$$

For $u = 15/16$, bounding the exponential-series tail geometrically after the quadratic term yields

$$e^{15/16} < 1 + u + \frac{u^2}{2} + \frac{u^3/6}{1 - u/4} = \frac{64133}{25088} < \frac{64}{25}. \tag{20}$$

Substitution into (18)–(19) leaves, respectively, the positive rational lower bounds

$$\frac{63}{500} \qquad \text{and} \qquad \frac{257931}{2048000}.$$

Hence

$$C_1(a) < \frac{4}{5}, \qquad C_2(a) < \frac{4}{5} \quad (0 < a \le 1/4). \tag{21}$$

From (17), $b < 32/5$. The constant trial satisfies $b(1 + a^2) < 34/5 < 8$, hence $R_0 < b$. Equations (13) and (21) give

$$R_1(b) < 2 + \frac{17}{25}(b - 2) < b,$$
$$R_2(b) < 6 + \frac{(b - 2)(b - 8)}{5} < 6 < b.$$

All three quotients are below $b \le \lambda_1$, which settles this subcase.

6.3 The complement for $a \ge 1/4$

Now use three radial-constant angular trials $e^{im\theta}$. Put

$$\alpha = \frac{2\log(1/a)}{1 - a^2}, \qquad y = \frac{b}{2\alpha}, \qquad \delta = \frac{1 + a^2}{8} - \frac{1}{4\alpha}. \tag{22}$$

Direct integration completes the square:

$$R_m = \alpha(m - y)^2 + b^2\delta. \tag{23}$$

Choose an integer nearest to y and its two neighbors. These are three distinct angular modes and their distances from y are at most $3/2$.

The number α is the average of $1/z$ over $[a^2, 1]$ and decreases with a. Since $e^{3/2} > 4$ already follows from the cubic Taylor sum,

$$\alpha \le \frac{32\log 4}{15} < \frac{16}{5}. \tag{24}$$

Also $\delta = (A - L)/4$, where A and L are the arithmetic and logarithmic means of 1 and a^2. The standard mean inequalities $a \le L \le A$ give

$$0 \le \delta \le \frac{(1 - a)^2}{8}. \tag{25}$$

Using (17) and $a \geq 1/4$, the largest of the three quotients in (23) is strictly less than

$$\frac{9}{2(1+a)^2} + \frac{9\alpha}{4} \leq \frac{72}{25} + \frac{36}{5} = \frac{252}{25} < 12. \tag{26}$$

On the other hand, (2.3), $a \geq 1/4$, and $\pi > 3$ imply $W(a) > 16 - 4 = 12$. The three trial quotients are therefore strictly below the Dirichlet ground state. Together with the flux-window and small-radius subcases, this completes the proof of Theorem 1.1.

7 Discussion, scope, and reproducibility

The proof separates three mechanisms. Below $b = 2$, the decisive point is monotonicity of a disk-type quotient under removal of the central region. Between $b = 2$ and $b = 6$, two exactly solvable disk modes remain effective after restriction. Above $b = 6$, either the annular flux window supplies three below-Landau angular modes, or its shortness forces bounds that are closed by moment control or transverse confinement. The interfaces $b = 2$, $b = 6$, and $a = 1/4$ are all included in the adjacent exact estimates.

No statement is made for a nonconcentric hole, mixed boundary conditions, an independently added harmonic flux, nonuniform fields, many-body effects, or experimental spectra. The inequality is non-strict in the theorem because the proof only needs $\mu_3 \leq \lambda_1$, although most variational branches yield strict trial bounds.

All identities used above are elementary and exact. A supplementary Python script independently checks the Barta identity, the two explicit $b = 2$ modes, the derivative factorization, the moment antiderivatives, the constant-mode completion, and the decisive rational inequalities. The script is a secondary reproducibility aid; the mathematical argument does not rely on floating-point spectra. Regeneration instructions and the tested software versions accompany the source bundle.

Statements and declarations

Data and code availability. No empirical data were generated. The exact symbolic checker and its regeneration instructions are included with the source package.

Use of generative AI. Generative-AI tools assisted with literature triage, exploration and adversarial checking of proof strategies, preparation of symbolic-checking code and the regime schematic, and drafting of the manuscript. Such tools are not authors. The proof is presented explicitly so that every claim can be checked independently; responsibility for any submitted version rests with the human author.

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